Supplementary material for 'Transient scaling and resurgence of chimera states in coupled Boolean phase oscillators'

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I. DERIVATION OF A PHASE MODEL FOR BOOLEAN PHASE OSCILLATORS

In this section, we derive a phase model for Boolean phase oscillators shown in the setup in Fig. 1(c) in the main article. The oscillators are described by output variables $\{x_i\}_{i=1}^N$, which are normalized output voltages $\{V_i\}_{i=1}^N$ of oscillators so that $x_i = 1$ ($x_i = -1$) corresponds to the high (low) Boolean voltage; specifically $x_i = 2V_i/V_H - 1$ with high Boolean voltage V_H [1]. The coupling of oscillators is realized with state-dependent delay that depends on the phase differences between the oscillator output x_i and K oscillatory input signals $\left\{x_{ji}^{\text{in}}\right\}$. Delay lines in the setup are based on chains of buffer gates that each add a gate propagation delay to the total delay as detailed in Ref. [1]. In the setup in Fig. 1(c), we include several delay lines of value $\sigma = 0.328 \pm 0.012\,\text{ns}$, corresponding to a single buffer gate and a long delay line of value $\tau = 8.4 \pm 0.3\,\text{ns}$, corresponding to 30 cascaded buffer gates. Several XOR logic gates are included as simple Boolean phase detectors [2] to generate a control signal $x_{ji}^c \in \{0, 1\}$. The control signal is given by a comparison of the output Boolean state of the oscillator x_i and an input Boolean state x_{ji}^{in}, according to

$$x_{ji}^c = x_{ji}^{\text{in}} \oplus x_i, \tag{S1}$$

where $\oplus : \{0, 1\} \times \{0, 1\} \to \{0, 1\}$ denotes the XOR Boolean function. Several multiplexers, which are 3-input logic gates, are included as Boolean switches to modify the feedback delay of the system by a delay σ_i depending on the control signals $\left\{x_{ji}^c\right\}$. This leads to the total feedback delay of the i-th oscillator according to

$$\tau_i = \tau_{m,i} - \sigma_i \sum_{j=0}^{K} x_{ji}^c, \tag{S2}$$

where $\tau_{m,i} = \tau_{0,i} + K\sigma_i$ denotes the maximum delay in the loop. This is a state-dependent delay because the $\left\{x_{ji}^c\right\}_j$ depend on the state of the local oscillator [3].

We also include a single inverter logic gate in the feedback loop in Fig. 1(c) that is known to lead to stable oscillations due to negative delayed feedback similar to a ring oscillator [4]. The resulting frequency of the i-th oscillator is, under the assumption of a constant feedback delay,

$$f_i = \frac{1}{2\tau_i}. \tag{S3}$$

Here the factor 2 comes in due to inverted delayed feedback, where two round trips lead to the initial state.

We assume that Eq. (S3) is valid for state-dependent delay when the frequency f_i is replaced with the instantaneous frequency $\dot{\phi}_i$. Using Eqs. (S1)-(S3) and a Taylor approximation with a maximum error of $\approx 3\,\mathrm{MHz}$ for our parameters, we find

$$\dot{\phi}_i = \omega_{0,i} + \tilde{\sigma}_i \sum_{j=j_0}^{j_K} x_j(\phi_j) \oplus x_i(\phi_i), \tag{S4}$$

with angular frequency $\omega_{0,i} = 2\pi/(2\tau_{0,i}) = 2\pi \cdot 9.3\,\mathrm{MHz}$ (for in-degree $K = 60$) and coupling strength $\tilde{\sigma}_i = 2\sigma\omega_{0,i}^2 = 0.089\,\mathrm{MHz}$ (using a value $\sigma = 0.515\,\mathrm{ns}$ to adjust for the error by the Taylor approximation). The summation from j_0 to j_K is over K coupling inputs corresponding to the non-local network topology. We replace $\{x_j\}$ with $\{\phi_j\}$ using the following definition of the phase ϕ_j from Ref. [5]

$$\sin[\phi_j(t)] = \frac{2x_j(t) - \max[x_j] - \min[x_j]}{\max[x_j] - \min[x_j]} = x_j(t), \tag{S5}$$

where x_j oscillates between the Boolean values -1 and 1. We express the XOR Boolean function with a difference of Heaviside functions and add a phase-lag parameter α_{ij}, leading to the following phase oscillator model

$$\dot{\phi}_i = \omega_{0,i} + \tilde{\sigma}_i \sum_{j=j_0}^{j_K} \left|\Theta[\sin(\phi_j)] - \Theta[\sin(\phi_i + \alpha_{ij})]\right|, \tag{S6}$$

The phase-lag parameter α_{ij} is vital for observing chimera states [6, 7] and is likely caused experimentally by delays along wire connections.

We couple the Boolean oscillators in a ring network with non-local coupling realized with a coupling radius R, corresponding to a rectangular coupling kernel. Specifically, the resulting dynamical equation for the network is

$$\dot{\phi}_i = \omega_{0,i} + \tilde{\sigma}_i \sum_{j=i-R}^{i+R} \left|\Theta[\sin(\phi_j)] - \Theta[\sin(\phi_i + \alpha_{ij})]\right|, \tag{S7}$$

where we consider a network of N nodes ($i \in \{1, 2, ..., N\}$), and periodic boundary conditions. The free-running frequency of an oscillator $\omega_{0,i}$ as a function of R results from the construction of the oscillator with logic gates as shown in Fig 1(c) in the main article. It is given by $\omega_{0,i} = 1/[2(30\tau_{\mathrm{buf}} + 2R\tau_{\mathrm{mux}} + 2R\tau_{\mathrm{buf}})]$ with $n_0 = 30$ the number of buffer gates used to build the delay line denoted τ, and the propagation delays of multiplexer and buffer are $\tau_{\mathrm{mux}} = 0.404 \pm 0.014\,\mathrm{ns}$, $\tau_{\mathrm{buf}} = 0.328 \pm 0.012\,\mathrm{ns}$, respectively.

FIG. S1: Frequency evolution from numerical simulation of Fig. S8 with parameters $N = 27$, others as in Fig. 4(c) in the main article. Initial condition and approximation as Fig. 5 in the main article.

For numerical simulation, it is advantageous to replace the Heaviside functions and the absolute value in Eq. (S7) with a smooth tanh, leading to

$$\dot{\phi}_i = \omega_{0,i} + \tilde{\sigma}_i \sum_{j=i-R}^{i+R} \{\tanh\left[-c \sin(\phi_j) \sin(\phi_i + \alpha)\right] + 1\} /2 \tag{S8}$$

with slope $c = 4$ used in the numerical simulation.

The numerical simulation of Eq. (S8) leads to a phase and frequency profile (snapshot) shown and discussed in Fig. 5 of the main article. Here, we also discuss the temporal evolution of the frequency that is shown in Fig. S1. Similar to Fig. 3(b) in the main article, the system shows a chimera state that wanders around the network (compare also the corresponding explanation in the main text). After a certain time that depends heavily on the initial conditions, the chimera state collapses to a synchronized state, which happens after 4.2 ms in Fig. S1.

Note that chimera states appear with Eq. (S7) for phase lag parameters in the vicinity of $\alpha_{ij} = 0.1$ in the model, which is different from the common value of $\alpha = \pi/2 - 0.18$ used by Abrams and Strogatz to achieve chimera states [7]. This may be due to the differences in the coupling function of the two models, cf. [8], where a systematic phase reduction has been performed. Note also that the experiment does not include self-coupling, but it is common to assume self-coupling in previous numerical studies of Kuramoto networks.

II. PHASE SNAPSHOTS FOR FIG. 3(C) OF THE MAIN ARTICLE

In this section, we discuss the experimentally measured temporal evolution of frequencies and phases in Fig. 3(c) of the main article, which we have identified as complex dynamics that alternates between chimera states and desynchronization. For the reader's convenience, we show the same illustration in Fig. S2(a) with eight time points marked as (b)-(i), corresponding to the phase snapshots in Fig. S2(b)-(i), respectively. In the first snapshot in Fig. S2(b), there exists a region, where the oscillators have the same phase to within the experimental phase resolution of $\Delta\phi = 0.25$. Outside the dotted lines, the oscillators are not phase-synchronized. Therefore, this state corresponds to both synchronization and desynchronization, which is a chimera state. This state exists for six periods, where the synchronized region drifts considerably in the network. In Fig. S2(c), we find an intermediate state, where the phases of the oscillators are distributed randomly so that we cannot identify a phase-synchronized region; hence, it is an unsynchronized state. In Fig. S2(d), a chimera-like state is reached again, with similar properties to the phase snapshot in Fig. S2(b). Therefore, the dynamics change within a timescale similar to the oscillation period from chimera-like states to unsynchronized states and back.

Figure S2(e) shows the phases of the oscillators in the network at a time when two regions (marked with dotted lines) show the same phases within the measurement error. Outside the two regions, the oscillators show large differences in the phase and are hence unsynchronized. This state corresponds to a chimera state of two synchronized and two desynchronized regions, which has been reported before with numerical simulations [8–11]. Similar to the findings in these references, the two synchronized cores are phase shifted by π.

Figures S2(f)-(i) show the phases of oscillators in the network in the region called VI in the main article. The synchronized region of the chimera state moves from the range $i \cong 75$ to $i \cong 120$ to the range $i \cong 115$ to $i \cong 20$ (periodic). In Fig. S2(h), one oscillator in the synchronized region does not synchronize, which is non-ideal.

FIG. S2: (a) Temporal evolution of the frequency averaged over 500 ns windows in the experiment, copied from Fig. 3(c) in the main article. Dotted lines mark the phase pictures shown in (b)-(i) at times $0.94\,\mu$s, $1.14\,\mu$s, $1.37\,\mu$s, $2.05\,\mu$s, $4.10\,\mu$s, $4.42\,\mu$s, $4.84\,\mu$s, and $5.26\,\mu$s, respectively. Experimental parameters as in Fig. 3 in the main article.

III. THE NEARLY SYNCHRONIZED STATE IN SIMULATION AND EXPERIMENT

In the main article, we find that the network synchronizes at high frequencies in the experiment, but at low frequencies, equal to the natural frequency, in the model [compare Figs. 3(a) in the main article and Fig. S1]. To get more insight into the synchronized state, we show the frequency and phase profiles in Fig. S3. We find that in the experiment, the phases are not synchronized and the frequencies are synchronized apart from two nodes that

FIG. S3: Snapshots of the synchronized state in (a) experiment and (b) numerical simulation of Eq. (S8) with $\alpha_{i,j} = 0.1 = \text{const}$ at (a) $t = 480\,\text{s}$ in transient of Fig. 3(a) in the main article and at (b) $t = 36\,\mu\text{s}$. The simulation was initialized in the synchronized state, but we confirmed with $N = 30$ that this state is also reached after a transient via chimera states. Other parameters as in Fig. 5 in the main article.

have $\sim 1\%$ higher frequency. In the model, on the other hand, both frequency and phase synchronize.

This difference between experiment and model could be, as discussed in the main text, due to heterogeneity in the phase lag parameter α_{ij} in the experiment. When heterogeneity is included in our model in Eq. (S7), however, the synchronized state is still almost synchronized with frequency differences below 0.01% for heterogeneity $\sigma_\alpha < 0.2$ ($\alpha_{i,j} = 0.1 \pm \sigma_\alpha$ according to a Gaussian distribution with standard deviation σ_α), while, for $\sigma_\alpha \geq 0.2$, a stable chimera state is found.

IV. DETERMINING THE FRACTION OF CHIMERA STATES IN THE TRANSIENT

As described in the main article, the transient dynamics includes chimera dynamics among other, more complex dynamics, with chimera-like states that exist only for a few oscillation periods alternating with an unsynchronized state. Here, we determine the fraction of chimera states in the transient with a measure similar to Ref. [12].

The chimera states that we observe in this study have a typical arch shape, where one domain of adjacent oscillators are synchronized at low frequency, and another domain is unsynchronized with higher frequencies as shown in Fig. 2(b) of the main article. Such a shape can be detected by measuring the standard deviation of frequencies

$$\sigma = \sqrt{\langle (f_i - \langle f_i \rangle)^2 \rangle}, \tag{S9}$$

and the average frequency difference between adjacent oscillators

$$\Delta = \langle |f_i - f_{i-1}| \rangle \,. \tag{S10}$$

An arch-shaped function has a large σ but a small Δ, so that a large

$$\psi = \sigma/\Delta, \tag{S11}$$

is a measure to detect chimera states.

We calculate this function for the partly synchronized temporal evolution in Fig. 3(c) in the main article, which we have copied for easier comparison in Fig. S4(a). Figure S4(b) shows ψ for this temporal evolution, exhibiting a peak above a value of $\psi = 1.5$ approximately when we detect chimera states in the main article. Therefore, we use a threshold of $\psi > 1.5$ to distinguish chimera states. With this numeric value, we find that chimera states appear for about 14% of the transient. This number is obtained from 100 measurements of length 6 μs on a network of $N = 128$ oscillators within the transient shown in Fig. 3(a) of the main article.

V. TECHNICAL DETAILS OF THE IMPLEMENTATION OF OSCILLATOR NETWORKS ON ELECTRONIC CHIPS

The experiments in the main article are realized with unclocked circuits of logic gates on the field-programmable gate array (FPGA) Altera Cyclone IV EP4CE115F29C7. The chip is separated into 7155 logic array blocks of 16 programmable logic elements of 4-inputs each. A Boolean phase oscillator is realized with up to 218 logic elements (depending on the frequency adjustment to reduce heterogeneity), requiring 14 logic array blocks. The layout of the oscillators on the chip is shown in Fig. S5. Allocated logic array blocks are marked in darker blue while unused logic array blocks are marked in light blue. The blue rectangle marks a region reserved for the dynamical network. The orange rectangle marks a region reserved for the readout and the acquisition of the network dynamics. About 25% of the resources on the FPGA are used for the largest implemented network of $N = 128$ nodes.

FIG. S4: (a) Copy of Fig. 3(c) in the main article. (b) Corresponding function ψ to detect a chimera state. Specifically, a chimera is detected from time 3.6 μs until 5.3 μs when $\psi > 1.5$ (shown with dashed lines).

This implementation does not account for the ring structure of the network, so that the delay between some neighboring oscillators, *e.g.*, 18 and 19 and especially 1 and 128, is long compared to other neighbors, such as 1 and 2. We estimate the maximum transmission delay between two oscillators to be 2 ns corresponding to the transmission delay across the chip , which is small compared to the period of about 100 ns.

In addition, the experiment implements heterogeneous wiring, where the j-th input x_{ji}^{in} of node i [Fig. 1(c) in the main article] is given by the j-th element of the sorted version of the list $\{(i-R) \bmod N, (i-R+1) \bmod N, ..., (i+R) \bmod N\}$. As a consequence, the phase adjustments are made at different positions in the ring depending on the oscillator index.

FIG. S5: Picture showing the arrangement of logic gates on the FPGA for the implementation of the network with $N = 128$ nodes. The picture was generated with the Altera Chip Planner.

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